Pseudocode

```
MODULE WelcomeMessage
        DISPLAY “Welcome to JeepTrack!”
END MODULE

MODULE DisplayLocations
        DISPLAY “Available Locations”
        DISPLAY “> San Pablo City                    > SM San Pablo”
        DISPLAY “> LSPU-San Pablo City Campus        > Liliw”
        DISPLAY “>Nagcarlan                          >Tiaong”
        DISPLAY “> Calauan                           >Los Banos”
        DISPLAY “> Sta Cruz”
END MODULE

MODULE PassengerTypes
        DISPLAY “Passenger Type”
        DISPLAY “> Regular                           > Student”
        DISPLAY “> PWD                               >Senior Citizen”
END MODULE

START
        CALL WelcomeMessage

        INITIALIZE locations as ARRAY OF STRING = {“SAN PABLO CITY”, “SM SAN PABLO”, “LSPU-SAN PABLO
        CITY CAMPUS”, “LILIW”,  “NAGCARLAN”, ”TIAONG”, “CALAUAN”, “LOS BANOS”, “STA CRUZ”}
        DECLARE currentLocation AS STRING
        DECLARE destination AS STRING

        WHILE true
                CALL DisplayLocation

                DISPLAY “Enter your current location (enter “exit” to quit): ”
                INPUT currentLocation

                INITIALIZE locationFound = false

                FOR EACH location IN locations ARRAY
                        IF (currentLocation == locations[i]) THEN
                                SET locationFound = true
                                EXIT loop
                        ELSE IF (currentLocation CONVERT TO LOWERCASE == “exit”) THEN
                                DISPLAY “Exiting the program…”
                                EXIT PROGRAM
                        END IF
                END FOR

                IF locationFound is true THEN
                        EXIT loop
                END IF

                DISPLAY “LOCATION NOT FOUND!”
        END WHILE

        WHILE true
                CALL DisplayLocation

                DISPLAY “Enter your destination (enter “exit” to quit): ”
                INPUT destination
```

```
        INITIALIZE locationFound = false


        FOR EACH location IN locations ARRAY
                IF (destination == locations[i]) THEN
                        SET locationFound = true
                        EXIT loop
                ELSE IF (destination CONVERT TO LOWERCASE == “exit”) THEN
                        DISPLAY “Exiting the program…”
                        EXIT PROGRAM
                END IF
        END FOR

        IF locationFound is true THEN
                EXIT loop
        END IF

        DISPLAY “LOCATION NOT FOUND!”
END WHILE

INITIALIZE routes = currentLocations + “-” + destination
DECLARE travelDistance AS FLOAT
DECLARE travelTime AS INTEGER

SWITCH routes
        CASE valid route
                SET travelDistance = route distance
                SET travelTime = route time
        DEFAULT:
                DISPLAY “ROUTE NOT FOUND!”
                EXIT PROGRAM
END SWITCH

INITIALIZE passengerTypes AS ARRAY OF STRING = {“REGULAR”, “SENIOR CITIZEN”, “PWD”,
“STUDENT”}
DECLARE passengerType AS STRING

WHILE true
        CALL PassengerType

        DISPLAY “Enter passenger type: ”
        INPUT passengerType

        INITIALIZE validPassenger = false

        FOR EACH passenger IN passengerTypes
                IF (passengerType == passengerTypes[i]) THEN
                        SET validPassenger = true
                        EXIT loop
                END IF
        END FOR

        IF validPassenger is true
                EXIT PROGRAM
        END IF

        DISPLAY “INVALID PASSENGER TYPE!”
END WHILE
```

```
INITIALIZE initialDistance = 4.00
INITIALIZE initialRate = 14.00
INITIALIZE succeedingRate  = 2.00
INITIALIZE discount = 0.20
DECLARE fare AS FLOAT
DECLARE discountedFare AS FLOAT
DECLARE finalFare AS FLOAT

CALCULATE fare = (travelDistance - initialDistance) * succeedingRate + initialRate

IF (passengerType == “REGULAR”) THEN
        SET discount = 0
        SET finalFare = fare
ELSE
        CALCULATE discountedFare = fare * discount
        CALCULATE finalFare = fare - discountedFare
END IF

DISPLAY “JEEPTRACK”
DISPLAY “ROUTE”
DISPLAY “DISTANCE                EST TIME TRAVEL”
DISPLAY “travelDistance km       travelTime minutes”
DISPLAY “PASSENGER TYPE”
DISPLAY “PASSENGER TYPE          DISCOUNT”
DISPLAY “passengerType           discount”
DISPLAY “FARE”
DISPLAY “BOUND TO: currentLocation > destination
DISPLAY “FARE AMOUNT:   ₱ finalFare”
DISPLAY “HAVE A SAFE TRAVEL!”
```